

OOB_shift_reaper + NAMU — Free-Run, Watched, and the God of Gods

Kalier · Nmapper · Discoverer — gods within a run, but NAMU rules them all

Author & designer: Angela Lopez Mendoza (@angelahack1) — Creator of Tlamatini **The design (Angela's):** these three agents **run FREE** for up to **OOB_shift_reaper** (default **3600 s**), watched the whole time (a hang inside the window only logs a banner; a hang past second 3601 is reaped). **But above them all stands NAMU** — when the Tlamatini process itself ends (Ctrl+C / shutdown / kill), the free-run window is void: **everybody is killed instantly, no matter how long they have run.**

1. The Hierarchy of Authority

Authority	Who / when	Power	Overrides
NAMU — the God of Gods (Tier-3)	the Tlamatini PROCESS ending: Ctrl+C (SIGINT), shutdown, SIGBREAK / SIGTERM, kill, atexit	when NAMU says STOP, EVERY running Kalier / Nmapper / Discoverer and every child they spawned is KILLED IMMEDIATELY — even a scan that started 30 s ago	EVERYTHING. The OOB window does not exist for NAMU.
OOB_shift_reaper (the free-run window)	during a NORMAL run	the three run FREE and watched; a hang inside 3600 s only logs a banner; a hang past 3601 is reaped	the per-agent wall-clock timeouts and the idle reaper

Rule of the realm: within a run the three agents are gods (free, immortal to timeouts). But **the gods die the instant NAMU dies**. A scan must never outlive Tlamatini.

2. OOB_shift_reaper — the three rules during a normal run

#	State of the agent	Elapsed time	What happens
1	WORKING — still producing output	any (even past 3600)	runs FREE. NEVER killed by OOB. It is a god.
2	HANGED — no output for <code>hang_detect_idle_seconds</code>	0 .. 3600 (inside the window)	a BANNER is logged ("possible hang, letting it run"). NOT killed. Keeps running.
3	HANGED — still doing nothing	second 3601 and beyond	KILLED effectively (tree-kill); partial output preserved; a KILL banner logged.

Within a run, the only OOB kill is rule 3. NAMU (Section 3) is the separate, supreme kill.

3. NAMU — the God of Gods (Tier-3)

When Tlamatini shuts down, the free-run window is void. NAMU hunts down and kills all three agents and their children, instantly.

Facet	Detail
WHO	NAMU = the Tlamatini process ending — Ctrl+C (SIGINT), a shutdown, SIGBREAK / SIGTERM, an outright kill, or normal <code>atexit</code> .
WHAT	Every running Kalier / Nmapper / Discoverer pool process and every scan child it spawned (<code>nmap.exe</code> , <code>nuclei.exe</code> , <code>subfinder.exe</code> , <code>naabu.exe</code> , <code>go.exe</code> , the in-flight HTTP/SSH call) is tree-killed IMMEDIATELY . No OOB grace. It does not matter if a scan has run 30 s or 30 min.
WHY	The OOB window makes these three immortal to timeouts DURING a run — but they must never outlive Tlamatini herself . A scan left running after the app is gone is an orphan (the exact <code>conhost.exe</code> / zombie problem the reaper exists to prevent).
WHERE	Registered in the Tlamatini CORE — <code>agent/apps.py:AgentConfig.ready()</code> already installs a shutdown / <code>atexit</code> / SIGINT / SIGBREAK cleanup path (the Tier-3 orphan reaper, <code>agent/orphan_reaper.py</code>). NAMU is an explicit, priority hunt inside that path.
HOW	On the shutdown signals, NAMU enumerates every live process whose cmdline references the three agents' pool dirs (<code>agents/pools/.../{kalier,nmapper,discoverer}</code> and <code>_chat_runs_{kalier,nmapper,discoverer}_*</code>) and <code>psutil-tree-kills</code> each — parent + all descendants — with a <code>taskkill /F /T</code> backstop. Immediate, unconditional, fail-safe (never raises into the shutdown path).
PRECEDENCE	NAMU fires BEFORE and ABOVE the OOB runner. The OOB runner's own rules (1-3) only ever apply while Tlamatini is alive; the moment NAMU is triggered, the OOB window is irrelevant and the kill is total.

NAMU is the God of Gods. When she says stop, everybody dies.

4. WHERE / WHY / HOW — per agent (the OOB free-run half)

4.1 Nmaper

WHERE	WHY	HOW
<code>nmaper.py:920</code> scan callsite; <code>command_timeout</code> (900) at <code>:845</code>	today hard-kills nmap at 900 s	swap to the streaming <code>_run_cmd_oob(...)</code> runner: nmap runs FREE to <code>OOB_shift_reaper</code> , watched; banner inside 3600, reap at 3601 if still hanged
<code>--host-timeout (:908-916)</code>	nmap's own graceful self-limit	stays OPTIONAL, user-set; never forced

4.2 Discoverer

WHERE	WHY	HOW
<code>_run_scan</code> callsite (<code>discoverer.py:981</code>); <code>command_timeout</code> (1800) at <code>:1044</code>	killed at 1800 s	swap to <code>_run_cmd_oob(...)</code> ; nuclei / naabu / httpx run FREE to <code>OOB_shift_reaper</code> , watched
meta calls (<code>go install</code> , bootstrap, validate)	setup, not scans	keep their own short timeouts

4.3 Kalier

WHERE	WHY	HOW
<code>kalier.py:489</code> <code>timeout</code> (300) then <code>urlopen(timeout=)</code> at <code>:514</code>	OUR HTTP side gives up at 300 s	the HTTP wait becomes <code>OOB_shift_reaper</code> ; Tlamatini waits FREE up to 3600 s for the Kali box
the Kali box <code>server.py</code> <code>COMMAND_TIMEOUT</code> (~180 s, external)	ONE honest fact: not ours	raising the box cap (or SSH) is what lets the remote scan itself reach 3600 s

4.4 The reaper / core

WHERE	WHY	HOW
<code>command_watchdog.py</code>	during a run the three are exempt	their children aren't shells, so already off its radar; the OOB runner's rule 3 is the only in-run kill
<code>apps.py</code> + <code>orphan_reaper.py</code> shutdown path	this is where NAMU lives	on shutdown, NAMU tree-kills all three agents + children immediately (Section 3)

5. Config Keys

Key	Default	Scope	Meaning
<code>OOB_shift_reaper</code>	3600	Kalier, Nmapper, Discoverer ONLY	free-run window; a hang is fatal only PAST this many seconds (rule 3)
<code>hang_detect_idle_seconds</code>	120	same three	no output for this long = "doing nothing (hanged)" — trigger for the banner and the rule-3 kill

NAMU has **no config knob** — she is unconditional. When Tlamatini ends, everybody dies, always.

6. What the Log Shows

Rule 2 (hang inside the window) — banner, keeps running:

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=====
POSSIBLE HANG: nmap has produced no output for 128s (elapsed 640s).
LETTING IT RUN FREE inside the 3600s OOB_shift_reaper window;
will only kill if still doing nothing past 3600s.
=====
```

Rule 3 (hang past the window) — reaped at 3601:

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OOB REAPER: nmap is doing nothing (no output for 210s) and is PAST the
3600s OOB_shift_reaper free-run window -> KILLING at 3601s.
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NAMU (Tlamatini shutting down) — everybody dies, regardless of elapsed:

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=====
NAMU — GOD OF GODS: Tlamatini is shutting down. Killing ALL running
Kali / Nmap / Discoverer scans and their children NOW, regardless
of the OOB_shift_reaper window. (a nmap scan had run only 30s.)
=====
```

7. What This Deliberately Does / Does NOT Do

- **DOES:** on shutdown (NAMU) kill all three + children instantly, no matter the elapsed time.
- Does NOT kill a working process during a run — ever.
- Does NOT kill a hanged process inside the OOB window — only logs a banner.
- Does NOT touch any agent other than these three.
- The only external limit acknowledged: the Kali box's own `COMMAND_TIMEOUT`.

Tlamatini — "one who knows." Design by Angela Lopez Mendoza. Within a run: `OOB_shift_reaper` = 3600 s, free / watched / reaped-if-hanged-past-3601. Above all: NAMU, the God of Gods — when Tlamatini stops, everybody is killed.